

UNIVERSITY ERP SYSTEM — SHORT REPORT

1. Introduction

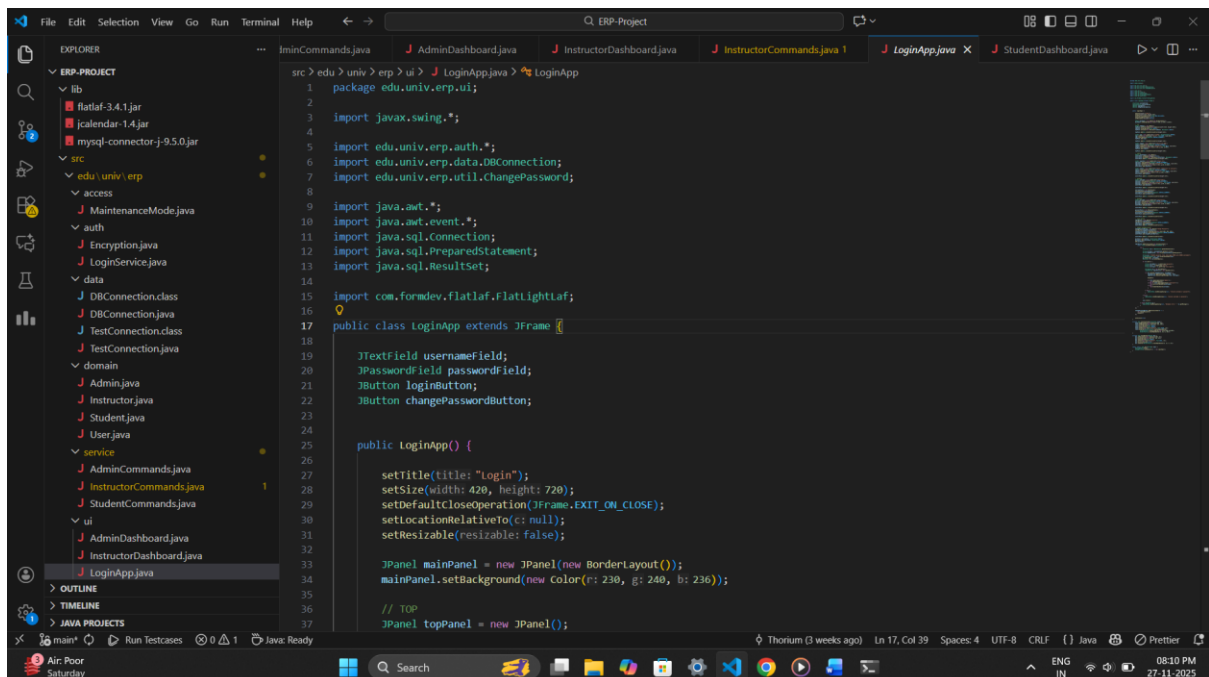
This project implements a complete University ERP System designed to manage academic operations for three main user roles:

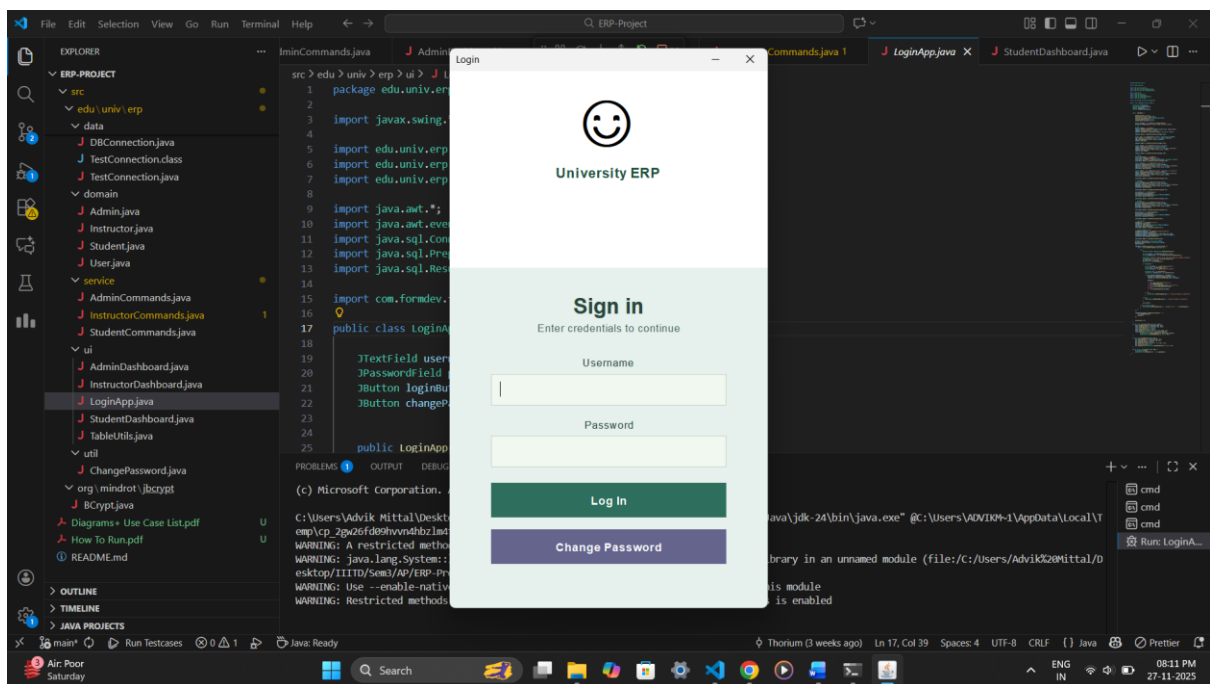
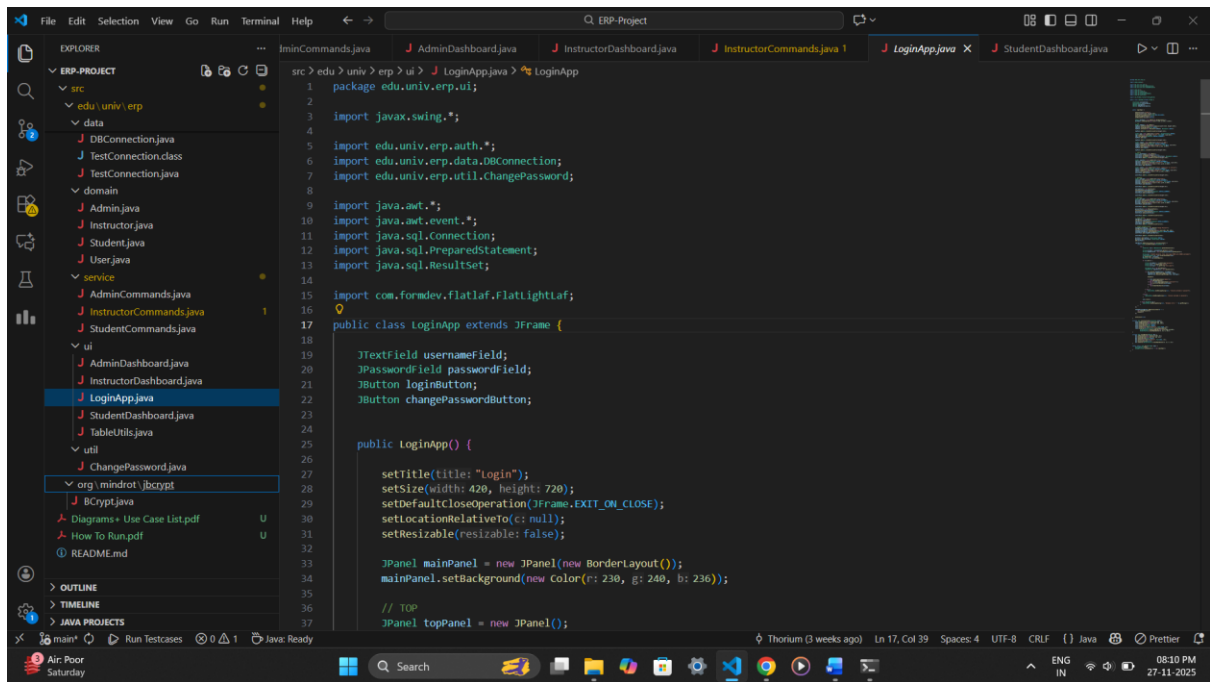
- Admin
- Instructor
- Student

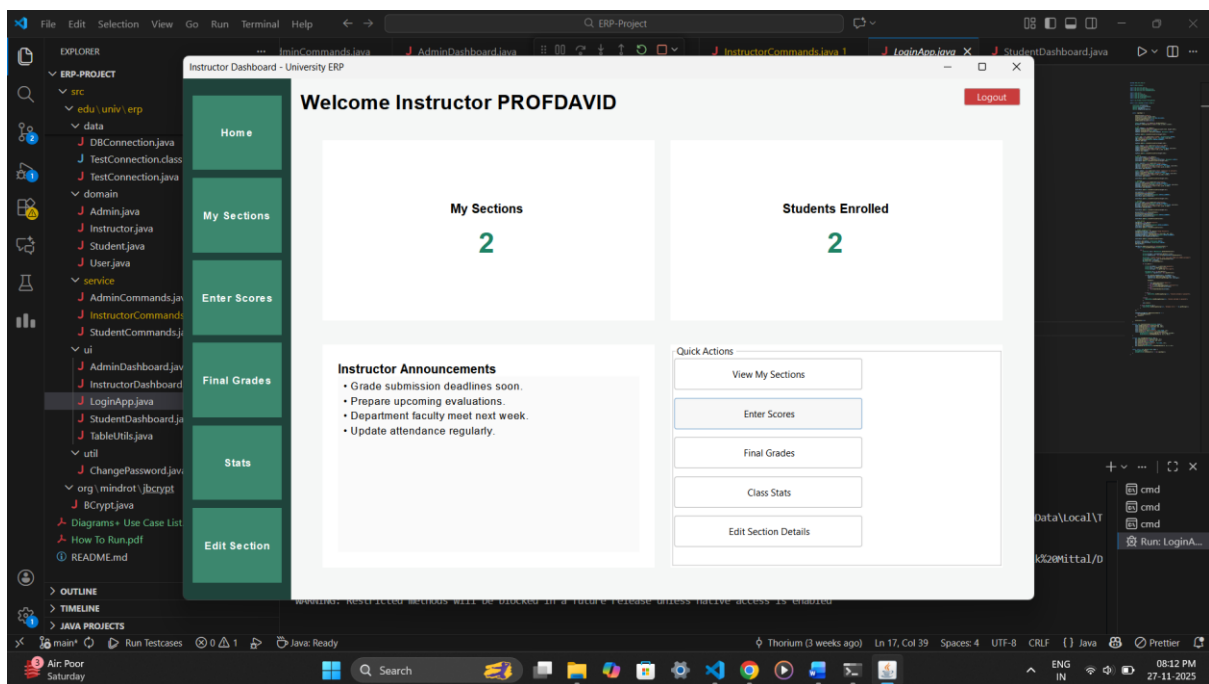
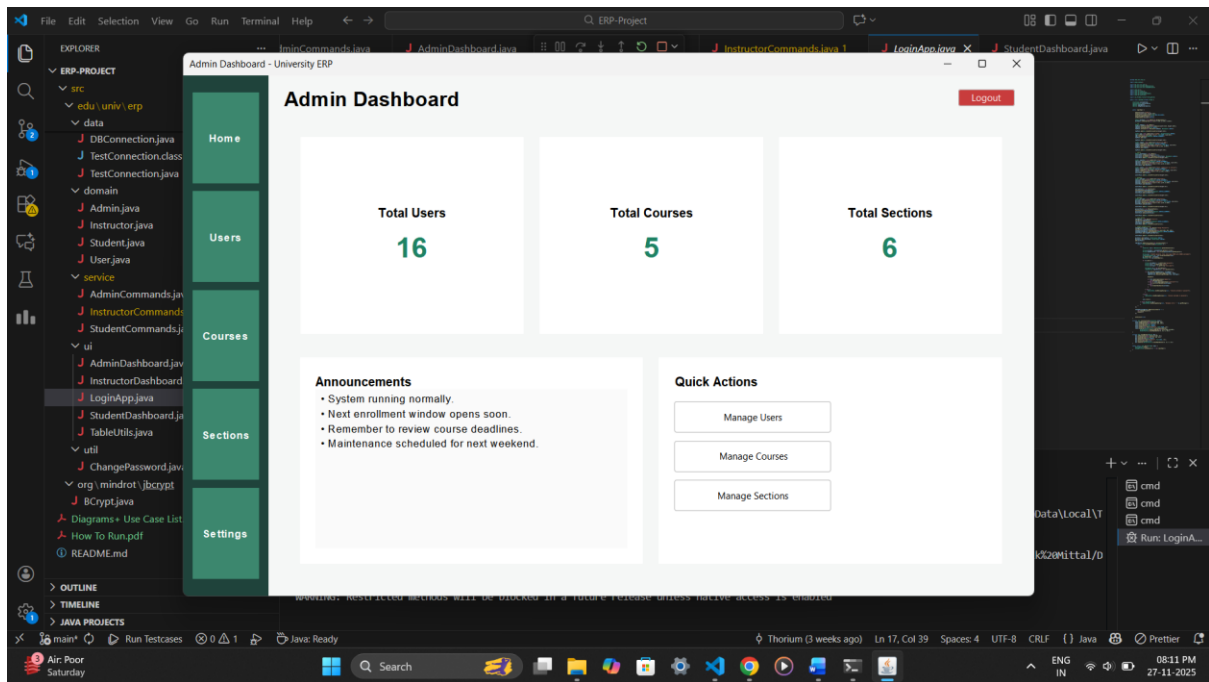
The application handles tasks such as course and section creation, student enrollment, score entry, grade computation, timetable display, and maintenance mode enforcement. It is built using Java Swing, MySQL, and follows an MVC-like separation between UI components, command classes, and database access.

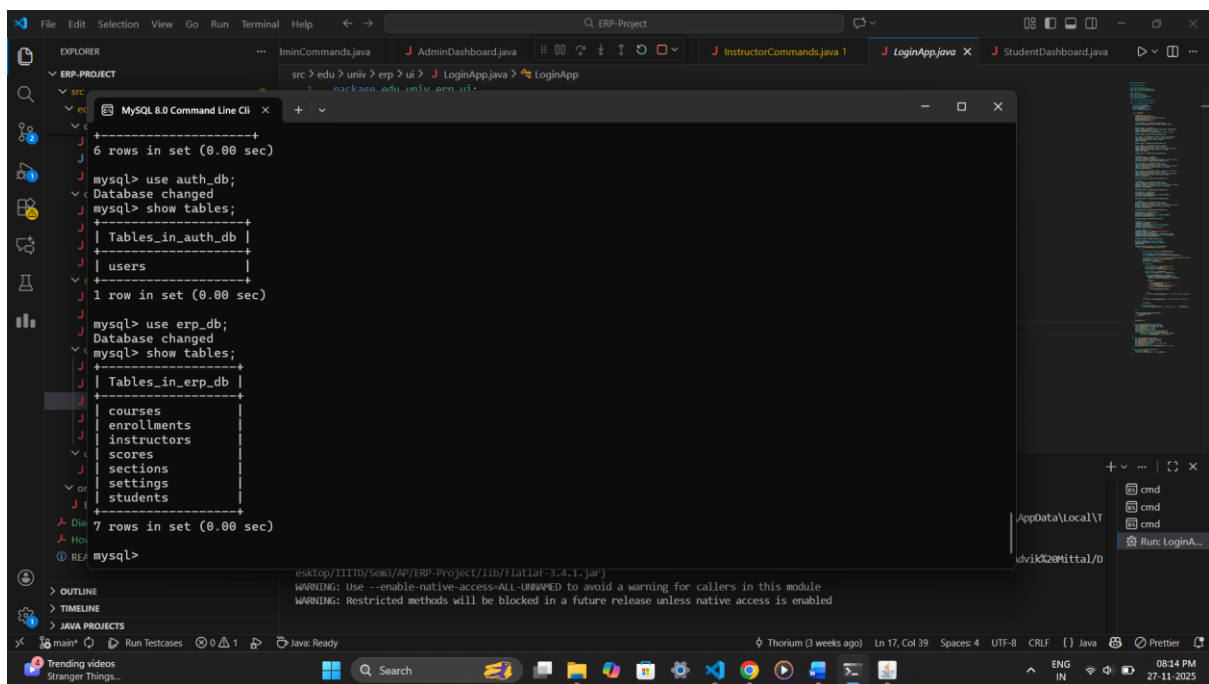
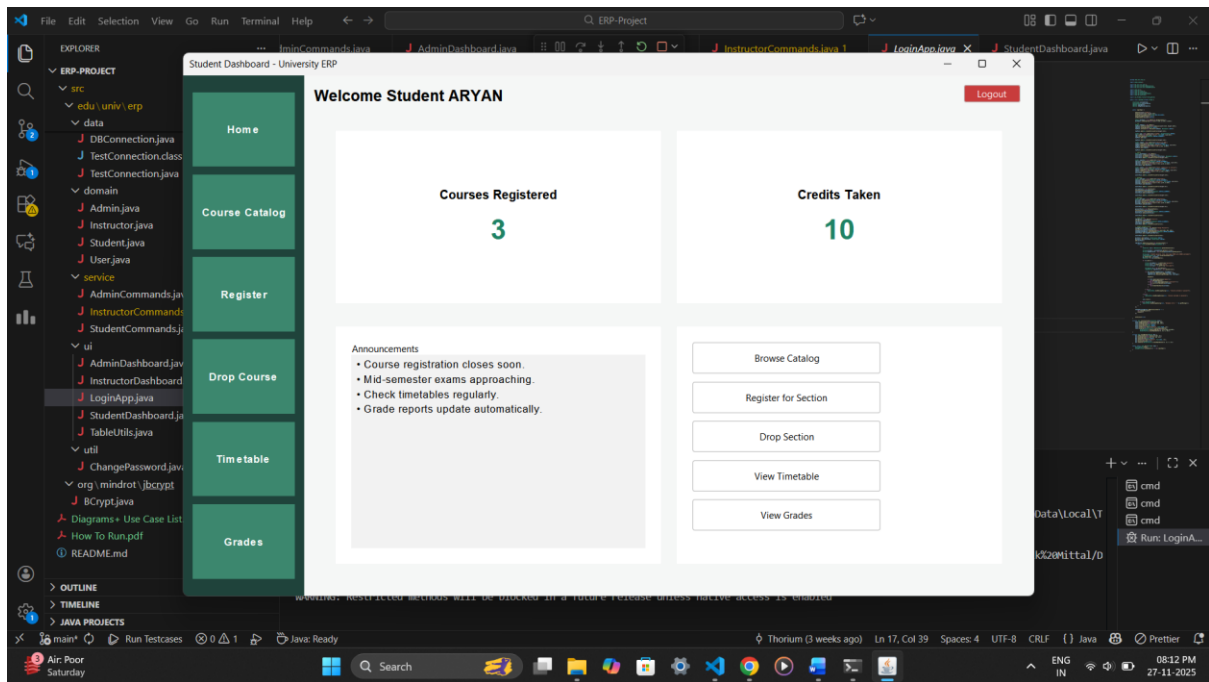
The goal of this system is to provide an intuitive, secure, and role-specific interface that automates essential university academic processes.

2. System Screenshots









3. Final-Grade Weighting Rule

The system uses a straightforward weighted evaluation model.

Each assessment entered by the instructor has:

- score: value from 0 to 100
- weight: percentage contribution (e.g., 20%)

Final grade for a student in a section is calculated using:

$$\text{Final Grade} = \sum \left(\text{score} \times \frac{\text{weight}}{100} \right)$$

Example:

Assessment Score Weight Contribution

Quiz 80 20% 16

Midterm 70 30% 21

Final Exam 90 50% 45

Total Final Grade = 82/100 (i.e 16+21+45)

The computation is implemented in:

- `InstructorCommands.computeFinalGrades(sectionId)`
- Student views this value in the Grades panel.

Validation enforced:

- Weight must be numeric
- Score must be between 0–100
- Section final grade is shown once per section

4. Role Enforcement

Role-based access is strictly implemented through:

- `auth_db.users(role)` column
- Dashboard switching after login
- Each dashboard shows features based on role

Admin

Allowed:

- User management (add Students/Instructors/Admins)
- Course creation
- Section creation
- Editing courses & sections

- Turning Maintenance Mode ON/OFF

Instructor

Allowed:

- View only their own sections
- View enrolled students in their sections
- Enter scores for students
- Compute & export final grades
- Edit only their own section's day_time/capacity
- View statistics (avg/min/max)

Blocked when Maintenance Mode is ON:

- Enter Scores
- Edit Section

Student

Allowed:

- Browse course catalog
- Register for available sections (capacity must be >0, deadline not passed)
- Drop sections (before course deadline)
- View timetable
- View grades
- Export grades to CSV

Blocked when Maintenance Mode is ON:

- Register
- Drop

These restrictions ensure system integrity and prevent unauthorized operations.

5. Maintenance Mode Enforcement

Maintenance Mode is stored in:

`maintenanceMode(isEnabled)`

Only Admin can toggle this.

When turned ON:

Blocked Actions

- Student:
 - Register for Section
 - Drop Section
- Instructor:
 - Enter Scores
 - Edit Section

Allowed Actions

- Viewing catalog
- Viewing grades
- Exporting CSV
- Viewing timetable
- Viewing final grades
- Viewing stats

UI Indicators

- Instructor Dashboard: red banner at top
 - Student Dashboard: red banner at top
 - Admin Dashboard: red banner at top
-

6. Database Table List

AUTH_DB

```
mysql> use auth_db;
Database changed
mysql>
mysql> SELECT
->     table_name,
->     column_name,
->     data_type,
->     is_nullable,
->     column_key,
->     column_default
-> FROM information_schema.columns
-> WHERE table_schema = 'auth_db';
```

TABLE_NAME	COLUMN_NAME	DATA_TYPE	IS_NULLABLE	COLUMN_KEY	COLUMN_DEFAULT
users	id	int	NO	PRI	NULL
users	username	varchar	NO		NULL
users	password	varchar	NO		NULL
users	role	varchar	NO		NULL
users	last_login	datetime	YES		NULL

5 rows in set (0.00 sec)

ERP_DB

```
->     data_type,
->     is_nullable,
->     column_key,
->     column_default
-> FROM information_schema.columns
-> WHERE table_schema = 'erp_db';
```

TABLE_NAME	COLUMN_NAME	DATA_TYPE	IS_NULLABLE	COLUMN_KEY	COLUMN_DEFAULT
courses	code	varchar	NO	PRI	NULL
courses	title	varchar	NO		NULL
courses	credits	int	NO		NULL
courses	deadline	date	YES		NULL
enrollments	enrollment_id	int	NO	PRI	NULL
enrollments	student_id	int	NO	MUL	NULL
enrollments	section_id	int	NO	MUL	NULL
enrollments	status	varchar	NO		NULL
instructors	user_id	int	NO	PRI	NULL
instructors	name	varchar	NO		NULL
instructors	department	varchar	NO		NULL
scores	section_id	int	NO	PRI	NULL
scores	student_id	int	NO	PRI	NULL
scores	assessment	varchar	NO	PRI	NULL
scores	score	double	NO		NULL
scores	weight	double	NO		NULL
sections	section_id	int	NO	PRI	NULL
sections	course_id	varchar	NO	MUL	NULL
sections	instructor_id	int	NO	MUL	NULL
sections	day_time	varchar	NO		NULL
sections	room	varchar	NO		NULL
sections	capacity	int	NO		NULL
sections	semester	varchar	NO		NULL
sections	year	int	NO		NULL
settings	key	varchar	NO	PRI	NULL
settings	value	varchar	NO		NULL
students	user_id	int	NO	PRI	NULL
students	roll_no	varchar	NO	UNI	NULL
students	program	varchar	NO		NULL
students	year	int	NO		NULL
students	name	varchar	YES		NULL

7. Conclusion

This ERP system implements all required features for managing university academic workflows across Admin, Instructor, and Student roles. With full support for grading, registration, capacity management, maintenance restrictions, and data export, it provides a functional and user-friendly application.

The project demonstrates:

- Real database integration
- Role-based access control
- UI design with Swing
- Correct enforcement of academic rules
- Robust error handling and validation

Overall, the system is complete, stable, and ready for demonstration and evaluation.